Generative Methods

Images

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Generative Methods

Outline

Generative vs. Discriminative

Generative Models for Images

Variational Autoencoder (VAE)

Generative Adversarial Networks (GANs)

Denoising diffusion (DDPM)

Summary

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1. Generative vs. Discriminative
2. Generative Models for Images
3. Variational Autoencoder (VAE)
4. Generative Adversarial Networks (GANs)
5. Denoising diffusion (DDPM)
6. Summary

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Generative vs. Discriminative

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Motivation

Why generative models?

Discriminative vs. generative

- **Discriminative:** learn $p(\boldsymbol{y} \mid \boldsymbol{x})$ or a decision boundary (e.g. classifiers).
- **Generative:** learn a distribution over data, typically $p(\boldsymbol{x})$ or $p(\boldsymbol{x}, \boldsymbol{y})$.

Why do we care about $p(x)$?

- **Sampling:** draw new realistic images, sentences, or tabular records.
- **Data augmentation:** synthesize extra examples for rare classes.
- **Imputation & denoising:** fill in missing values, remove noise.
- **Compression / representation learning:** learn useful latent codes.

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Probability view

Joint, conditional, and marginal

- Suppose $(\boldsymbol{x}, y)$ with input $\boldsymbol{x}$ and label y.
- **Joint:** $p(\boldsymbol{x}, y)$, **marginal:** $p(\boldsymbol{x}) = \sum_y p(\boldsymbol{x}, y)$.
- **Conditionals:**

$$p(y \mid \boldsymbol{x}) = \frac{p(\boldsymbol{x}, y)}{p(\boldsymbol{x})}, \qquad p(\boldsymbol{x} \mid y) = \frac{p(\boldsymbol{x}, y)}{p(y)}.$$

- Generative models can be used for classification via

$$p(y \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid y)\, p(y),$$

but in modern deep learning we often focus on generation and representation.

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Images as data

What do we want to generate?

- Image $\boldsymbol{x}$ can be seen as a vector in $\mathbb{R}^{H \times W \times C}$.
- We want to learn $p(\boldsymbol{x})$ such that samples $\tilde{\boldsymbol{x}} \sim p$:
  - look realistic to humans,
  - cover the main modes of the data distribution (no trivial memorization),
  - can be controlled (e.g. conditional on class label).
- Typical datasets: MNIST, Fashion-MNIST, CIFAR-10, CelebA, ImageNet (hard).

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Families of image generators

High-level landscape

- **Autoencoders (AE):** encoder $\boldsymbol{x} \mapsto \boldsymbol{z}$, decoder $\boldsymbol{z} \mapsto \hat{\boldsymbol{x}}$ (typically a **deterministic** bottleneck).
- **Variational Autoencoders (VAE):** probabilistic AE with latent prior $p(\boldsymbol{z})$ and ELBO training.
- **Generative Adversarial Networks (GAN):** generator $G(\boldsymbol{z})$ vs. discriminator $D(\boldsymbol{x})$ in a minimax game.
- **Diffusion / score-based models:** gradually add noise then learn to reverse the process.
- **Autoregressive models:** factorize $p(\boldsymbol{x})$ pixel-by-pixel (PixelCNN, PixelRNN).

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MNIST demos: repository layout

samples/ (course code)

Paths (relative to the Chapter 9 slides folder)

- `samples/vae/` — Variational Autoencoder on flattened MNIST (`train_mnist_vae.py`, `config.yaml`).
- `samples/gan/` — GAN with **MLP** or **DCGAN**-style conv (`train_mnist_gan.py`; `config.yaml` vs. `config_dcgan.yaml`).
- `samples/diffusion/` — DDPM with a small U-Net (`train_mnist_diffusion.py`, `config.yaml`).

Common pattern

Each folder: `requirements.txt`, YAML config, script, and a `results/` directory with figures + `results.txt` (metrics and paths).

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Data protocol

Train / validation / test (MNIST)

- Official MNIST **training** set (60,000 images) is split into:
  - **train** fraction = `train_ratio` (e.g. 0.8–0.9) for optimization,
  - **validation** = remainder, for model selection (best checkpoint).
- Official MNIST **test** set (10,000) is held out for **final** evaluation only (same spirit as the RNN MNIST demo).
- All three demos normalize pixels consistently where needed (e.g. VAE: $[0, 1]$; GAN / diffusion: often $[-1, 1]$ for Tanh or Gaussian noise).

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Outputs you should inspect

What to show in class

Demo	Typical artifacts under `results/`
VAE	`samples.png` (prior samples), `reconstructions.png` (input vs. reconstruction), `loss_curve.png`, `results.txt`
GAN	`samples.png` (generator grid), `loss_curve.png` (D/G losses), checkpoints `best_G.pt` / `best_D.pt`
Diffusion	`samples.png` (reverse-process samples), `loss_curve.png` (noise MSE), `checkpoints/best_ddpm.pt`

Side-by-side: VAE (often softer) vs. GAN (sharper but unstable) vs. diffusion (sharpness vs. compute trade-off).

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Variational Autoencoder (VAE)

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VAE

What we want (big picture)

- Data $\boldsymbol{x}$ is high-dimensional, but $p_{\text{data}}(\boldsymbol{x})$ lies on a **thin manifold**.
- Goals:
  - **Generate** new samples $\boldsymbol{x}$
  - Learn a **compact latent** representation $\boldsymbol{z}$
- Key idea:
  - Introduce latent $\boldsymbol{z}$ with simple prior $p(\boldsymbol{z})$ (e.g. $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$)
  - Learn how data are generated via $\boldsymbol{z} \to \boldsymbol{x}$

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VAE

Autoencoder (AE)

- Deterministic mapping:

$$\boldsymbol{z} = f(\boldsymbol{x}), \quad \hat{\boldsymbol{x}} = g(\boldsymbol{z})$$

- Learns compression / reconstruction
- **No explicit generative model**
- Cannot sample $\boldsymbol{z}$ to generate new data

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VAE

Variational Autoencoder (VAE)

- Encoder outputs a **distribution**:

$$q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$$

- Sample $\boldsymbol{z} \sim q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$
- Decoder:

$$p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$$

- **Generative path:**

$$\boldsymbol{z} \sim p(\boldsymbol{z}) \rightarrow \boldsymbol{x} \sim p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$$

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VAE

Key difference

AE

- Deterministic mapping $\boldsymbol{x} \to \boldsymbol{z}$
- No explicit generative story

VAE

- Probabilistic encoder $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$
- Can sample $\boldsymbol{z} \sim p(\boldsymbol{z})$ to generate

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VAE

Generative model (conceptual view)

Ingredients:

- **Prior** $p(\boldsymbol{z})$ (often $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$) — where unconditional samples come from.
- **Decoder** $p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$ — how $\boldsymbol{x}$ is generated from $\boldsymbol{z}$.
- **Encoder (training only)** $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ — approximate posterior; we do not need it for pure generation after training.

Pipeline (training): $\boldsymbol{x} \xrightarrow{\text{encoder}} q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \xrightarrow{\text{sample}} \boldsymbol{z} \xrightarrow{\text{decoder}} \hat{\boldsymbol{x}}$.

- $\boldsymbol{z}$ is **random**, not a single deterministic code — we fit and regularize the whole family $q_\phi(\cdot \mid \boldsymbol{x})$.

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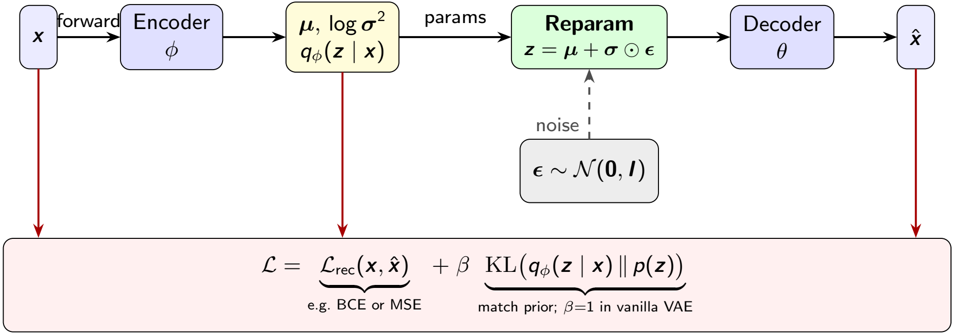
VAE

Training graph (computational view)



- **Trainable:** ϕ and θ appear on *differentiable* paths; sampling z is non-differentiable, but **reparameterization** routes gradients w.r.t. ϕ through (μ, σ).
- **Objective:** minimize $\mathcal{L}$; β controls reconstruction vs. regularization (standard ELBO uses $\beta = 1$).

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VAE

Using the model: reconstruct, generate, and encode

- **Reconstruct:** $\boldsymbol{x} \to$ encoder $\to \boldsymbol{z}$ (often $\boldsymbol{z} = \boldsymbol{\mu}$) $\to$ decoder $\to \hat{\boldsymbol{x}}$; or sample $\boldsymbol{z} \sim q_\phi$ for stochastic reconstructions.
- **Generate:** $\boldsymbol{z} \sim p(\boldsymbol{z})$, then $\hat{\boldsymbol{x}}$ from decoder only — **no encoder**.
- **Encode / latents:** $\boldsymbol{x} \mapsto \boldsymbol{z}$ via encoder for visualization, features, or downstream tasks.

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VAE

Two terms in words (before heavy math)

- **Reconstruction:** make $\hat{\boldsymbol{x}}$ close to $\boldsymbol{x}$ (in code: often BCE with sigmoid pixels, or MSE under a Gaussian decoder).
- **Regularization:** keep $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ close to prior $p(\boldsymbol{z})$ so latents stay in a region where sampling works — implemented as **KL divergence**.
- **Trade-off:** fidelity vs. structured latent space (often weighted by β in β-VAE).
- **Reparameterization:** sample $\boldsymbol{z}$ so gradients reach encoder parameters (derived in the mathematical view).

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VAE on MNIST

samples/vae/ — tie ideas to code

What the script implements

- Flatten 28×28; MLP encoder $\rightarrow (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$; $\boldsymbol{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \epsilon$; MLP decoder $\rightarrow \hat{\boldsymbol{x}}$ with **sigmoid**; loss = reconstruction (BCE-style) + KL (Gauss–Gauss).
- `config.yaml`: `latent_dim`, `hidden_dims`, `beta`, `epochs`, `train_ratio`; train/val on 60k, test on 10k.

Inspect

`reconstructions.png`, `samples.png`, `loss_curve.png`, `results.txt`.

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Why VAE samples often look "soft"

Beyond "train longer"

- Factorized likelihood / ELBO can favor **averages** across plausible shapes $\Rightarrow$ **blurry** edges.
- More epochs help only partly; perceptual sharpness is not the main selling point vs. GANs / diffusion.

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VAE

Step 0: what we cannot compute directly

Generative model: prior $p(\boldsymbol{z})$, decoder $p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$.

1. **Marginal density / evidence:** $p_\theta(\boldsymbol{x}) = \int p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})\, \mathrm{d}\boldsymbol{z}$. For fixed $\boldsymbol{x}$, as a function of θ, the same expression is the **marginal likelihood** $L(\theta)$ (often loosely called the **likelihood** in MLE). High-dimensional integral $\Rightarrow$ no closed form for flexible $p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$.
2. **True posterior:** $p_\theta(\boldsymbol{z} \mid \boldsymbol{x}) = \dfrac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})}{p_\theta(\boldsymbol{x})}$. Unknown $p_\theta(\boldsymbol{x})$ in the denominator $\Rightarrow$ posterior also intractable.
3. **Variational idea:** introduce an *approximate* posterior $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ (encoder) and reason with Kullback–Leibler divergences instead.

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VAE

Step 1: importance identity (true equality)

Fix data $\boldsymbol{x}$. Choose $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ that covers all relevant latent regions (so we do not divide by zero where $p_\theta(\boldsymbol{x} \mid \boldsymbol{z})p(\boldsymbol{z})$ is nonzero).

$$
\begin{aligned}
p_\theta(\boldsymbol{x}) &= \int p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})\, \mathrm{d}\boldsymbol{z} && \text{marginal: integrate out } \boldsymbol{z} \text{ under } p(\boldsymbol{z}) \\
&= \int \frac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\, q_\phi(\boldsymbol{z} \mid \boldsymbol{x})\, \mathrm{d}\boldsymbol{z} && \text{multiply \& divide by } q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \\
&= \mathbb{E}_{\boldsymbol{z} \sim q_\phi(\cdot \mid \boldsymbol{x})}\left[\frac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\right] && \int (\cdot)\, q_\phi\, \mathrm{d}\boldsymbol{z} = \mathbb{E}_{q_\phi}[\cdot]
\end{aligned}
$$

Take log:

$$
\log p_\theta(\boldsymbol{x}) = \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\right].
$$

Still hard: log remains *outside* the expectation.

VAE

Step 2: Jensen $\Rightarrow$ a tractable lower bound

For any positive random variable Y, log is **concave**, so (Jensen)

$$\log \mathbb{E}[Y] \;\geq\; \mathbb{E}[\log Y].$$

Apply with $Y(\boldsymbol{z}) = \dfrac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}$:

$$\log p_\theta(\boldsymbol{x}) \;\geq\; \mathbb{E}_{q_\phi}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) + \log p(\boldsymbol{z}) - \log q_\phi(\boldsymbol{z} \mid \boldsymbol{x})] =: \mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}).$$

This is the **ELBO** (evidence lower bound). Group terms:

$$\mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}) = \mathbb{E}_{q_\phi}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})] - \mathbb{E}_{q_\phi}\left[\log \frac{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}{p(\boldsymbol{z})}\right].$$

By definition $\mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p(\boldsymbol{z})) = \mathbb{E}_{q_\phi}\left[\log \frac{q_\phi(\boldsymbol{z}|\boldsymbol{x})}{p(\boldsymbol{z})}\right]$, so

$\mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}) = \mathbb{E}_{q_\phi}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})] - \mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p(\boldsymbol{z}))$.

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VAE

Step 3: read ELBO term-by-term

$$\mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}) = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})]}_{\text{(A) fit data after decoding}} - \underbrace{\mathrm{KL}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p(\boldsymbol{z})\big)}_{\text{(B) keep latent posterior close to prior}} .$$

1. **(A) Reconstruction/likelihood term:** encourages decoder parameters θ to make $\boldsymbol{x}$ probable from sampled $\boldsymbol{z}$.
2. **(B) KL term:** regularizes encoder output so latent codes stay in regions where sampling from $p(\boldsymbol{z})$ is meaningful.
3. **Optimization target:** maximize ELBO (equivalently minimize $-\mathcal{L}_{\mathrm{ELBO}}$) over both θ and ϕ.

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Step 4: why ELBO is a lower bound

Why this KL appears: we already introduced an approximate posterior $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ for the true posterior $p_\theta(\boldsymbol{z} \mid \boldsymbol{x})$. So a natural "gap" measure is $\mathrm{KL}(q_\phi \| p_\theta(\cdot \mid \boldsymbol{x}))$. Using Bayes: $p_\theta(\boldsymbol{z} \mid \boldsymbol{x}) = \dfrac{p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) p(\boldsymbol{z})}{p_\theta(\boldsymbol{x})}$. **Now derive the exact gap (fix one data point $\boldsymbol{x}$):**

$$\begin{aligned}
\mathrm{KL}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \| p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big) &= \mathbb{E}_{q_\phi}\left[\log \frac{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}{p_\theta(\boldsymbol{z} \mid \boldsymbol{x})}\right] \\
&= \mathbb{E}_{q_\phi}[\log q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) - \log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) - \log p(\boldsymbol{z}) + \log p_\theta(\boldsymbol{x})] \\
&= \log p_\theta(\boldsymbol{x}) - \mathbb{E}_{q_\phi}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})] + \mathbb{E}_{q_\phi}[\log q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) - \log p(\boldsymbol{z})] \\
&= \log p_\theta(\boldsymbol{x}) - \mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}).
\end{aligned}$$

Rearrange:

$$\log p_\theta(\boldsymbol{x}) = \mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{x}) + \mathrm{KL}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \| p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big).$$

Since $\mathrm{KL} \geq 0$, ELBO is a lower bound; equality when $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) = p_\theta(\boldsymbol{z} \mid \boldsymbol{x})$.

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Key insight (bridge to training)

- From Step 4: $\log p_\theta(\boldsymbol{x}) = \mathcal{L}_{\text{ELBO}}(\boldsymbol{x}) + \text{KL}(q_\phi \| p_\theta(\cdot \mid \boldsymbol{x}))$ — *marginal log-likelihood* equals ELBO plus **posterior mismatch**.
- So maximizing ELBO pushes up a tractable lower bound while encouraging q_ϕ to track $p_\theta(\boldsymbol{z} \mid \boldsymbol{x})$.
- Joint training fits **both** the generative model p_θ (decoder) and the inference model q_ϕ (encoder).

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VAE

Step 5: from ELBO to trainable loss

1. **Sampling trick (reparameterization):**

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \quad \boldsymbol{z} = \boldsymbol{\mu}_\phi(\boldsymbol{x}) + \boldsymbol{\sigma}_\phi(\boldsymbol{x}) \odot \boldsymbol{\epsilon}.$$

This writes random $\boldsymbol{z}$ as a differentiable transform of (μ, σ) and noise.

2. **Mini-batch objective (negative ELBO, one MC sample):**

$$\mathcal{J}(\theta, \phi; \boldsymbol{x}) \approx -\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) + \mathrm{KL}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p(\boldsymbol{z})\big).$$

In code, $-\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$ is implemented as a reconstruction loss (typically BCE for Bernoulli decoder, or MSE for Gaussian decoder).

3. **β-VAE variant:**

$$\mathcal{J}_\beta \approx -\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) + \beta\, \mathrm{KL}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p(\boldsymbol{z})\big), \quad \beta > 0.$$

Larger β gives stronger latent regularization.

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VAE

ELBO roadmap (step-by-step recap)

1. Start from intractable $p_\theta(\boldsymbol{x})$ and $p_\theta(\boldsymbol{z} \mid \boldsymbol{x})$.
2. Introduce $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ and rewrite $p_\theta(\boldsymbol{x})$ as an expectation under q_ϕ.
3. Apply Jensen to obtain ELBO.
4. Split ELBO into likelihood fit $-$ KL regularization.
5. Use reparameterization to estimate gradients and optimize with SGD/Adam.

One-line memory: *fit data in observation space, regularize structure in latent space.*

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GANs

What we want (big picture)

- Learn a direct sampling map from noise to data:

$$\boldsymbol{z} \sim p(\boldsymbol{z}), \qquad \tilde{\boldsymbol{x}} = G_\theta(\boldsymbol{z}).$$

- No explicit encoder and no tractable likelihood $p_\theta(\boldsymbol{x})$ as in VAE.
- Training signal comes from a learned critic/classifier D_ϕ that distinguishes real vs. fake.

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GANs

Two networks, one game

1. **Generator** G_θ**:** synthesis path only, $\boldsymbol{z} \mapsto \tilde{\boldsymbol{x}}$.
2. **Discriminator** D_ϕ**:** outputs a realness score for an input image.
3. **Competition:** D tries to separate real/fake; G tries to fool D.

Intuition: as D improves, it provides better gradients; as G improves, fake samples move toward p_{data}.

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GANs

GAN training diagram

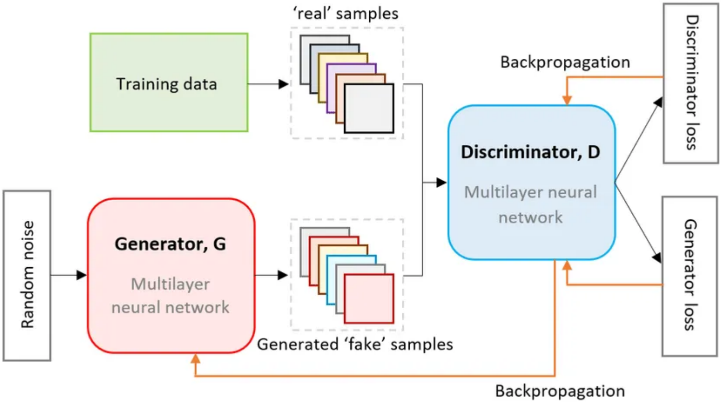

Figure 4.1: GAN training flow: noise → generator G; training data vs. generated fakes → discriminator D; separate losses and backpropagation to D and G. Source: Presidio, Exploring the Power of Generative Adversarial Networks (GANs) with Azure.

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GANs

Using the model after training

- **Generation only:** sample $\boldsymbol{z} \sim p(\boldsymbol{z})$, output $\tilde{\boldsymbol{x}} = G_\theta(\boldsymbol{z})$.
- **No encoder in vanilla GAN:** unlike VAE, there is no direct $\boldsymbol{x} \mapsto \boldsymbol{z}$ path.
- **No discriminator needed at inference:** D is mainly a training-time teacher.

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GANs

What code actually optimizes

1. **BCE with logits** for D (stable); often alternate k steps on D vs. one on G.
2. **Non-saturating** G**:** maximize $\mathbb{E}[\log D(G(\boldsymbol{z}))]$ instead of $\mathbb{E}[\log(1 - D(G(\boldsymbol{z})))]$ to avoid vanishing gradients.
3. **Label smoothing** on real targets (e.g. 0.9 instead of 1) to reduce overconfident D.
4. **Balance:** if D is too strong $\Rightarrow$ weak gradients for G; tune `lr_d`, `lr_g`, capacities, update ratio.

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GAN on MNIST (course code)

samples/gan/ — tie configs to outputs

Architectures

- `config.yaml` (`arch: mlp`): G, D MLP on flattened 784-D, Tanh pixels in $[-1, 1]$ — fast baseline; may need more tuning.
- `config_dcgan.yaml` (`arch: dcgan`): conv-transpose G + conv D on $1 \times 28 \times 28$; use e.g. `results_dcgan/` to compare side-by-side.

What to inspect

- Selection: val loss on D (real/fake BCE combined) — simple scalar, **not** FID.
- Artifacts: `best_G.pt`, `best_D.pt`; grid `samples.png`; curves `loss_curve.png`.

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GANs

Step 0: minimax objective (original)

$$\min_{\theta} \max_{\phi} \; \mathbb{E}_{\boldsymbol{x} \sim p_{\text{data}}}[\log D_\phi(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z} \sim p(\boldsymbol{z})}[\log(1 - D_\phi(G_\theta(\boldsymbol{z})))].$$

- For fixed G, the best D approximates a density-ratio classifier between real and generated data.
- At equilibrium, G pushes generated distribution toward p_{data} (in the Jensen–Shannon sense for the vanilla objective).

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GANs

Step 1: practical generator objective

- Original G term $\mathbb{E}[\log(1 - D(G(\boldsymbol{z})))]$ can saturate when D is very confident.
- Practical replacement (non-saturating):

$$\max_{\theta} \mathbb{E}_{\boldsymbol{z} \sim p(\boldsymbol{z})}[\log D_{\phi}(G_{\theta}(\boldsymbol{z}))].$$

- Same fixed point intuition, usually better gradients early in training.

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GANs

Roadmap + trade-offs

1. Sample $\boldsymbol{z}$; synthesize $\tilde{\boldsymbol{x}} = G(\boldsymbol{z})$; train D on real vs. fake.
2. Optimize minimax (or practical BCE / non-saturating variants).
3. At inference: **only** G — one forward pass per sample.

- **Pros:** often sharp samples; cheap generation after training.
- **Cons:** unstable training, mode collapse; no tractable $p_\theta(\boldsymbol{x})$; many variants (WGAN, StyleGAN,...).

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DDPM

What we want (big picture)

- Learn to generate data by **reversing noise**.
- Instead of one-shot generation (GAN), DDPM uses a **multi-step refinement** process.
- Core intuition:
  1. Corrupt real images gradually with Gaussian noise until nearly pure noise.
  2. Train a neural network to predict and remove noise at each step.
  3. At inference, start from random noise and denoise step-by-step to obtain a sample.

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DDPM

Why diffusion is attractive

- **Stable training objective:** supervised noise prediction (MSE-style), no adversarial minimax game.
- **High sample quality:** often competitive visual fidelity.
- **Trade-off:** generation is slower because sampling needs many reverse steps (T evaluations).
- Practical lens for students:
  - GAN = fast generation, harder training.
  - Diffusion = easier training, slower generation.

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DDPM for beginners

Minimum you must remember

1. **Forward noising:** gradually add Gaussian noise to clean image $\boldsymbol{x}_0$ until almost pure noise $\boldsymbol{x}_T$.
2. **Train target:** given $(\boldsymbol{x}_t, t)$, the model predicts the injected noise ϵ.
3. **Loss:** simple regression loss (typically MSE/L2) between true noise and predicted noise.
4. **Generation:** start from $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, denoise step-by-step to get $\boldsymbol{x}_0$.
5. **Trade-off:** usually stable and high quality, but sampling is slower than one-shot generators.

One-line memory

DDPM = "add noise with a known rule" + "learn to remove it step-by-step."

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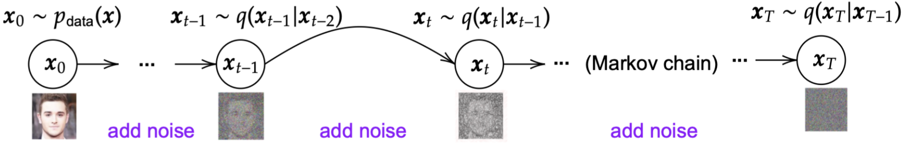
DDPM forward process

From $\mathbf{x}_0$ to $\mathbf{x}_t$ with key formulas

Forward

$\boldsymbol{x}_T \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$ when: T is large.



β_t : *small variance*; $\alpha_t = 1 - \beta_t$

$\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \qquad t \in [1, T] \subset \mathbb{N}^+$

$$\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\sqrt{\overline{\alpha}_t}\, \boldsymbol{x}_0,\ (1 - \overline{\alpha}_t)\mathbf{I}\right)$$

$$\boldsymbol{x}_t = \sqrt{\overline{\alpha}_t}\boldsymbol{x}_0 + \sqrt{(1 - \overline{\alpha}_t)}\boldsymbol{\epsilon}_t$$

Can compute x_t directly from x_0

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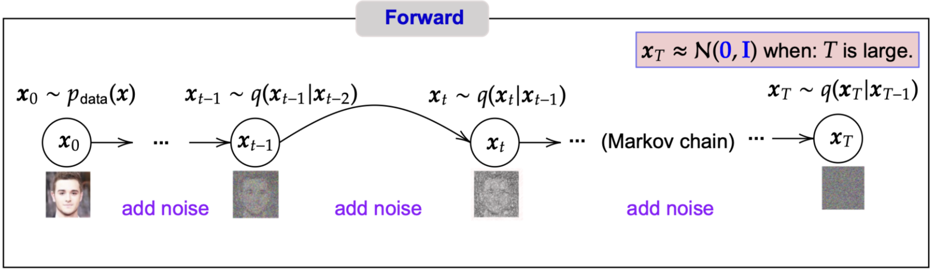
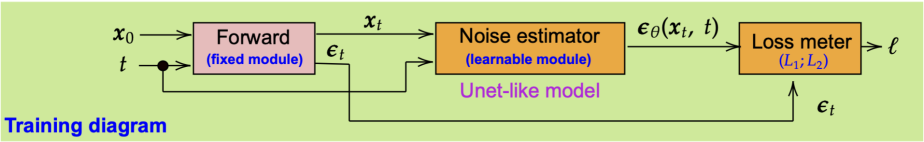
DDPM training view

Forward module + learnable noise estimator





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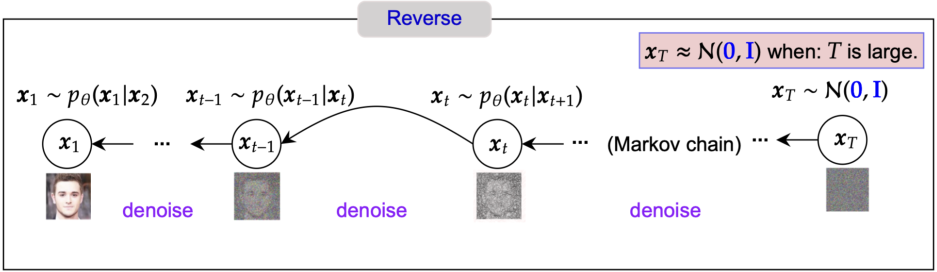
DDPM reverse process

Sampling by iterative denoising



Algorithm 2 Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
end for
return $\mathbf{x}_0$

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DDPM

Training/inference pipeline (no heavy math yet)

Training loop

1. Sample clean image $\boldsymbol{x}_0$ and random timestep $t \in \{1, \ldots, T\}$.
2. Add noise to obtain $\boldsymbol{x}_t$.
3. Predict noise $\hat{\epsilon} = \epsilon_\theta(\boldsymbol{x}_t, t)$.
4. Minimize prediction error against true ϵ.

Generation loop

Start from $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, then iteratively apply learned reverse updates $\boldsymbol{x}_T \to \boldsymbol{x}_{T-1} \to \cdots \to \boldsymbol{x}_0$.

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DDPM on MNIST (course code)

samples/diffusion/

Implementation choices

- Images scaled to $[-1, 1]$; linear β schedule from `beta_start` to `beta_end` over `timesteps` T.
- Small **U-Net** with residual blocks + **sinusoidal time embedding** + downsampling to 7×7 and upsampling with skip connections.
- Train/val/test: same MNIST split as other demos; best checkpoint by **val MSE** on noise prediction; test MSE reported in `results.txt`.
- **Cost:** sampling runs T network evaluations $\Rightarrow$ slower than one-shot $G(\boldsymbol{z})$ in GANs.

Classroom usage

If students are weak in math, focus on this computational recipe first (and skip detailed derivations in the next subsection).

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DDPM

Math roadmap (optional)

- Step 1: define forward Gaussian noising process.
- Step 2: derive closed-form $q(\boldsymbol{x}_t \mid \boldsymbol{x}_0)$.
- Step 3: parameterize reverse process with $\epsilon_\theta(\boldsymbol{x}_t, t)$.
- Step 4: use simplified training loss $\mathcal{L}_{\text{simple}}$.

Teaching note

Students can skip proof details and remember only: "*sample t, add noise, predict noise*".

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DDPM

Step 1–2: forward process and closed-form

- Fix a horizon T and a variance schedule $\beta_1, \ldots, \beta_T$ (small positive).
- Define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.
- Forward process (Gaussian transitions):

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\big(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1},\, \beta_t \boldsymbol{I}\big), \qquad q(\boldsymbol{x}_T \mid \boldsymbol{x}_0) \approx \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}).$$

- Closed-form at any t:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0,\, (1-\bar{\alpha}_t)\boldsymbol{I}\big).$$

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DDPM

Step 3–4: reverse process and objective

- Learn a network $\epsilon_\theta(\boldsymbol{x}_t, t)$ to predict the **noise** ϵ in

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}).$$

- Training loss (simple DDPM):

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\boldsymbol{x}_0,\epsilon}\left\|\epsilon - \epsilon_\theta(\boldsymbol{x}_t, t)\right\|_2^2.$$

- Sampling: start from $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, run T learned reverse steps (Markov Gaussian updates) to obtain $\boldsymbol{x}_0$.

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DDPM advanced math

Core equations (optional)

1) Forward transition

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1},\, \beta_t \boldsymbol{I}\right), \quad \alpha_t = 1 - \beta_t,\ \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s.$$

2) Closed-form noising at any step

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0,\, (1-\bar{\alpha}_t)\boldsymbol{I}\right), \quad \boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}).$$

3) Practical training loss

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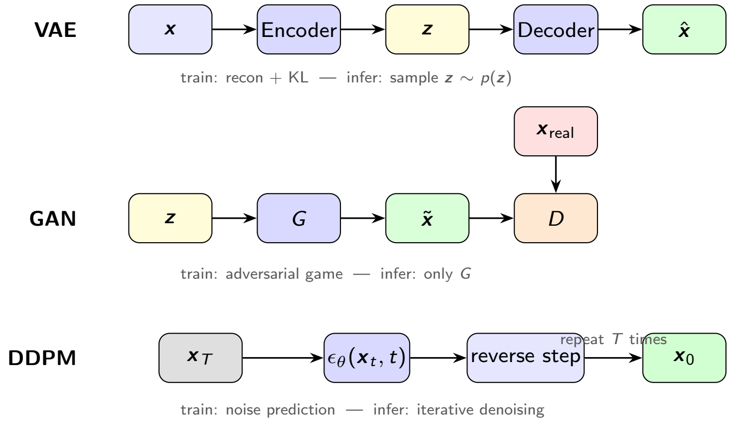
Pipeline comparison

VAE vs. GAN vs. DDPM (at a glance)



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Diffusion vs. VAE vs. GAN

Qualitative comparison (MNIST demos)

	VAE	GAN	DDPM
Likelihood	ELBO / approximate	no explicit $p(\boldsymbol{x})$	sequence of Gaussian steps
Sharpness	often softer	often sharper (if stable)	can be sharp; depends on T, schedule, capacity
Training	relatively stable	can be unstable	stable but slow sampling

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Summary

Key takeaways

Concepts

- Generative models aim to learn $p(\boldsymbol{x})$ or $p(\boldsymbol{x}, \boldsymbol{y})$, enabling sampling and data synthesis.
- **Images:** VAE (ELBO, often blurry samples), GAN (adversarial game, sharp but fragile), DDPM (noise prediction + iterative sampling).
- **Course alignment (images):** runnable MNIST baselines in `samples/vae`, `samples/gan` (MLP + DCGAN configs), `samples/diffusion` with YAML configs and `results/` artifacts for lectures.
- **Text generation** (tokenizers, causal LM training, decoding, KV-cache, code samples): covered in `slides/10-modern-llms/ModernLLMs.tex` — see pointers at the end of the diffusion section in this deck.
- Conditional generation and data synthesis have many applications but also raise ethical and societal concerns.

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